

To: Potential Homebuyers

Date: October 31, 2025

Subject: Predictive Modeling Analysis for Ames Housing Market

Introduction

The goal of this project is to develop an explanatory and progressive regression model to estimate home sale prices in Ames, Iowa. Using the Iowa housing dataset, our objective is two-fold: (1) identify key housing characteristics that significantly influence sale prices and (2) build a parsimonious model that balances interpretability and predictive accuracy.

The response variable in our analysis is SalePrice, the recorded sale price of each home (in U.S dollars). The variable represents the final transaction value and serves as the dependent variable we aim to predict based on various structural and property attributes.

Process

Preparing the Models

Our first order of business was to eliminate variables that were not useful for determining sale price. This led us to eliminate the PID, as it is an identifier and not an actual predictor. After that we looked at the scatterplots for each predictor. This helped us to weed out variables that had noticeable outliers or other glaring issues. One of the variables we eliminated was Lot_area, which had multiple influential outliers. The next step in our data cleaning was to remove all of the categorical variables. This included variables such as both of the building type variables and central air. Our final effort was to try and eliminate some of the predictors that had very similar or repeated meanings. Therefore we excluded options such as Halfbath, Fullbath, and Totalbath.

Making the Models

After completing this process, we moved on to making our models. Some of the methods we used to make our models include, backwards, forwards, stepwise, and personal selection. The processes used for backwards and forwards different in method, but produced the same results. For backwards selection (Model A), the p-value threshold was set to 0.05, we then used the step function to eliminate predictors until the 6 most significant predictors were remaining. At that point the 6 predictors of Overall_Qual, Year_Built, First_FlrSF, GRLivArea, Fireplaces, and GarageArea were used to make the model. Forward selection (Model B) had a similar process. However, instead of eliminating from a large pool of predictors like backwards selection, forwards instead chooses the best predictors first as you add steps. This led us to create a model with the same predictors chosen for the backwards selection.

As for stepwise selection (Model C), we started by choosing two predictors that we believed to be strong in the model, and then we fixed those predictors before we began a stepwise selection process. After we chose and locked in Overall_Qual and Year_Remod_Add, we then began the stepwise selection process to add the next best 4 predictors. This process added First_FlrSF, GRLivArea, Fireplaces, and GarageArea.

When it comes to the models made from personal selection, (Models D and E), the process was less uniform. For these two models it was left up to the user to determine which predictors to add to the model. Some factors that led to the choice in predictors include p-values, the data dictionary, and the users' own previous knowledge and experiences.

Selecting the Model

In order to choose our best model out of options A-E we used the common criteria used for comparing models. Some of the elements we compared were adjusted R^2 , mean square error, and p-values/ test statistics. Using these values we determined that our best option was model A. Model A has the highest adjusted R^2 , 0.7793, of any of our models, meaning that it best accounts for the variability displayed in sales price. It also had the lowest RMSE, 36460.5, meaning its predictions were closer on average to the actual sales prices. Lastly, this model had six predictors that all had p-values < 0.001 , which fits the stated requirements and means that the predictors are all significant to the model. These criteria made it clear to us that Model A, which was found using forwards selection, is our best model when it comes to predicting sale price.

Assumptions & Adding a Categorical Variable to the Model

Key Assumptions for Analysis:

- **Linearity:** All scatterplots (Graph A) of each predictor show a moderately strong linear relationship with response variable sales price. Although there are some outliers in the scatterplots, and the scatterplot for fireplaces that has discrete values displays a weaker linear relationship, all of the scatterplots and lines of fit have significant p-values, which means that they all have a significant linear relationship with sales price.
- **Independence:** The data comes from a random sample of all houses in the Ames housing market. Therefore, the data points are independent of each other.
- **Equal Variance:** In the residual plot (Graph B), outside of the outliers further from the cluster of predicted sales price data (x-axis), the residuals are nearly equally distributed above and below the line. Thus, the equal variance condition is met.
- **Normality:** The normal quantile plot (Graph C) shows, ignoring the tails, the quantiles of the residuals follow the line. This means that the data from the multiple regression model are approximately normally distributed.

All of the conditions for the regression model (Model A) are satisfied. Therefore, we will proceed with the Model A without making any transformations.

Adding a categorical variable to the model

We tested the model with an additional variable: building type. Before we added it to the model, we recoded the variable from five to three building types by combining “duplex” and “2famCon” houses into “2Fam”, and combining “Twnhs” and “TwnhsE” into “Twnhs”. After recoding the building type variable, we added it to the regression. In order to add the categorical variable into the model, we had to use dummy variables. We made single family homes the base variable, and used 2Fam and Twnhs as our dummy variables:

2Fam = 1 (if the building type is a 2Fam) or 0 (otherwise)

Twnhs = 1 (if the building type is a Twnhs) or 0 (otherwise)

With building types in the model, the adjusted R^2 increased and MSE decreased (Model F), the multiple regression model improved compared to Model A with 6 quantitative variables.

Results

Our best model, Model F, uses the following formula:

Predicted Sale Price = $-867,731.10 + 20,976.90 * \text{Overall_Qual} + 417.01 * \text{Year_Built} + 27.53 * \text{First_FlrSF} + 39.93 * \text{GRLivArea} + 10,118.33 * \text{Fireplaces} + 44.58 * \text{GarageArea} - 19,970.24 * \text{Bldg_Type 2_2Fam} - 15,395.74 * \text{Bldg_Type 2_Twnhs}$

The coefficients of each variable are to be interpreted as followed:

- Y-intercept: When all other predictors equal 0, the predicted sale price is $-\$867,731.10$, which is not possible and therefore not meaningful.
- Overall_Qual: A 1 unit increase in overall quality is predicted to increase the sale price by $\$20,976.90$ when all other predictors are held constant.
- Year_Built: For every 1 year increase in the year the house was built, the predicted sale price increases by $\$417.01$ when all other predictors are held constant.
- First_FlrSF: When first floor area increases by 1 square foot, the predicted sale price increases by $\$27.53$ when all other predictors are held constant.
- GRLivArea: When above grade living area increases by 1 square foot, the predicted sale price increases by $\$39.93$ when all other predictors are held constant.
- Fireplaces: When the number of fireplaces increases by 1 fireplace, the predicted sale price increases by $\$10,118.33$ when all other predictors are held constant.
- GarageArea: When garage area increases by 1 square foot, the predicted sale price increases by $\$44.58$ when all other predictors are held constant.
- Bldg_Type 2_2Fam: With all other predictors held constant, the predicted sale price for a 2 family home will be $\$19,970.24$ less than a single family home
- Bldg_Type 2_Twnhs: With all other predictors held constant, the predicted sale price for a townhouse will be $\$15,395.74$ less than a single family home

To test our regression model against a real-life house, we selected 3322 Harrison Rd, a relatively expensive home in a nice neighborhood of Ames, Iowa (please view Image A in the appendix).

We used real data from Beacon, a property records database covering Ames, to determine characteristics about the predictors for the value of this home. When determining the values of the predictors based upon this dataset, we had to make a few assumptions to fit reality into the simplified predictor values. The first assumption was the overall quality, which we assigned at 7 (“good”) due to the house being listed as “Normal” condition. The single family house was built in 2004 and is a single family home. We were able to find that the home has 1 gas fireplace, a gross living area (above grade) of 3,379 square feet, and a garage area of 848 square feet.

However, we were unable to find exact details about the square footage solely on the first floor. Likewise, we estimated that the first floor occupies approximately 60% of the gross square footage, giving us 2,027 square feet as the estimated first floor living area.

Inputting all of these values and estimates into our regression formula, we produced a predicted sale price for the home of \$353,444.19. The most recent appraisal in 2025 estimates the value of the house to be \$589,500 with a land value of \$114,000 for a total property value of \$703,500. This means that the model underpredicted by \$350,055.81.

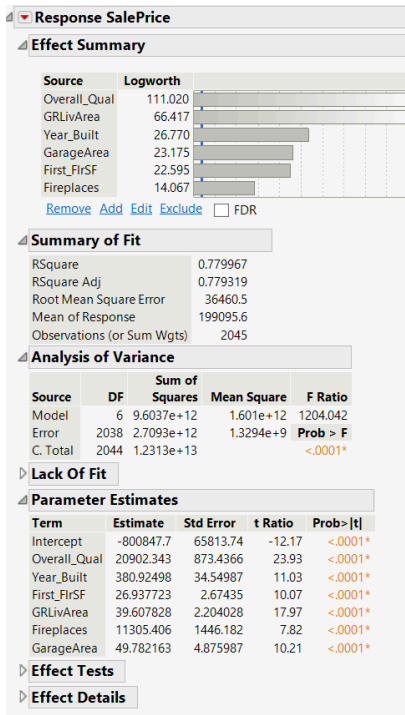
The 95% confidence interval for the mean is (346,232.26, 360,692.79), meaning that we are 95% confident the true average selling price of the homes in Ames is in the interval \$346,232.26 – \$360,692.79. A 95% prediction interval for the sale price of this home is (282,628.54, 424,296.50), meaning that we are 95% certain a home with identical characteristics to 3322 Harrison Rd would sell for \$282,628.54 – \$424,296.50. Please review Image B for the JMP table displaying both of these results.

Discussion

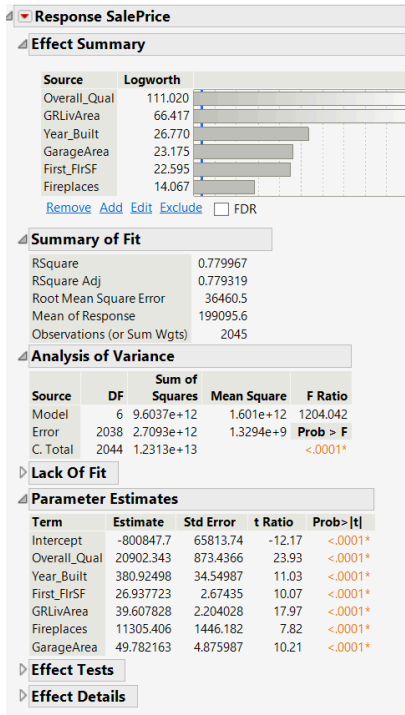
While our final regression model (Model F) demonstrates strong explanatory power—with a high adjusted R^2 and all predictors statistically significant—the model's real-world predictive accuracy is limited when applied to individual cases. Using the property at 3322 Harrison Rd as a test case, our model predicted a sale price of \$350,800.78, whereas the actual appraised value was \$703,500. This results in a residual of -\$352,699.22, indicating substantial underprediction. Although the 95% prediction interval \$282,628.54 – \$424,296.50 captures a reasonable range of expected variation, the true sale price falls far outside this interval, confirming that our model performs poorly for high-end properties. This discrepancy suggests that while the model fits the general housing market in Ames, it may not account for non-linear effects or variables that strongly influence luxury homes—such as lot location, neighborhood prestige, or recent renovations. In short, the model accurately captures the average trend and relationships in the dataset but lacks precision at the extremes, particularly for properties that deviate from the mean in quality or scale. Future refinements could include interaction terms, transformations, or additional categorical predictors to improve performance across all housing tiers.

Appendix

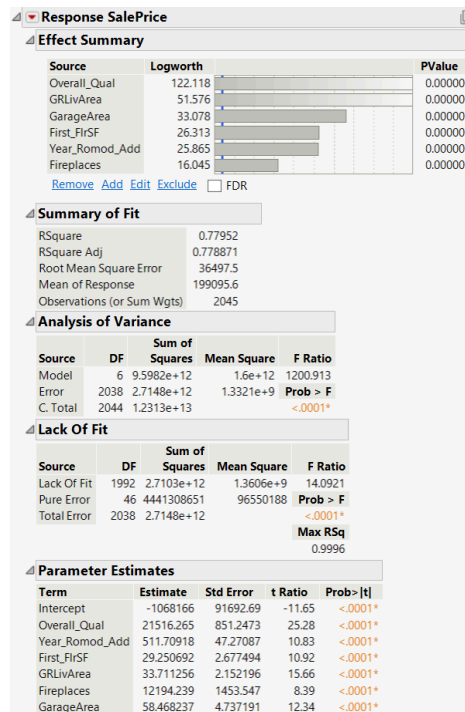
Model A: Forward Selection



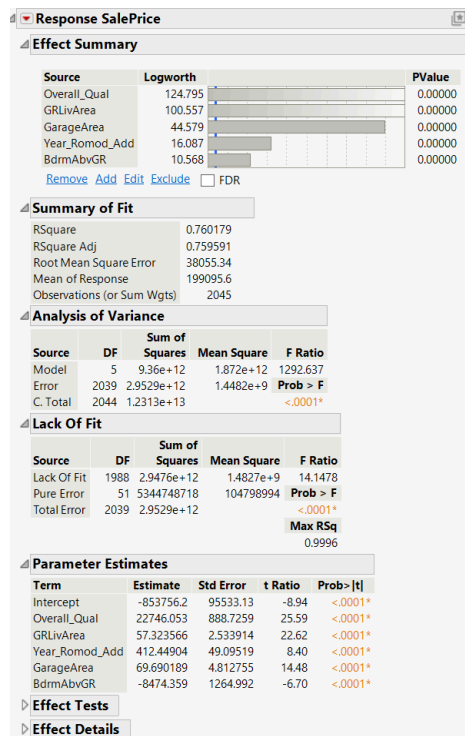
Model B: Backwards Selection



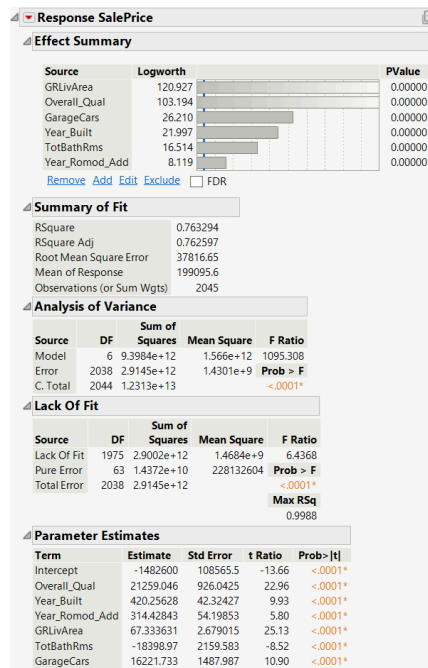
Model C: Stepwise Selection



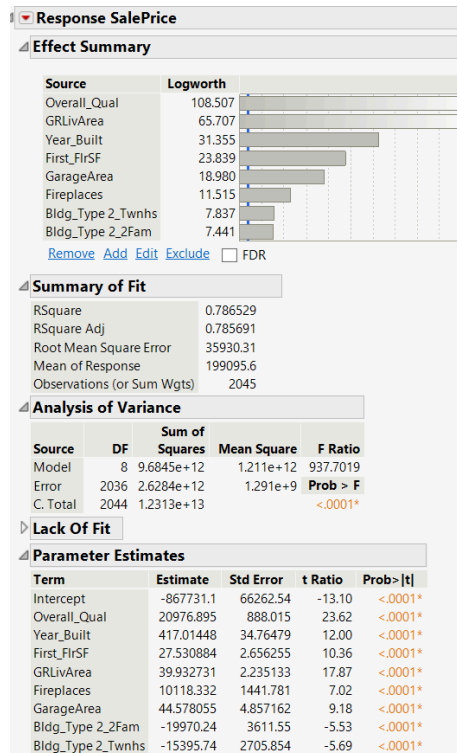
Model D: 5 regressors chosen based on logical assumption



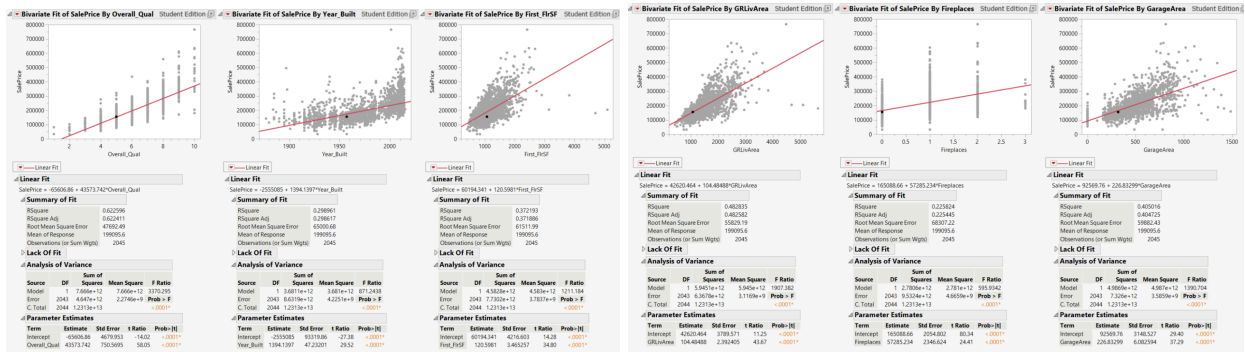
Model E: 6 regressors chosen based on each regressor's individual R^2



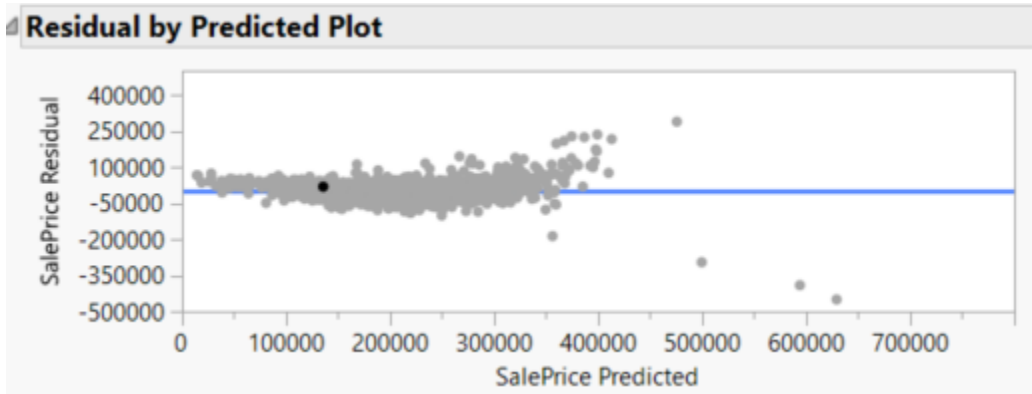
Model F: Dummy variable building type added into regression model (Model A)



Graph A: Individual scatter plots of the 6 predictors in Model A



Graph B: Residual plot of the multiple regression model (Model A)



Graph C: Normal quantile plot of the multiple regression model (Model A)

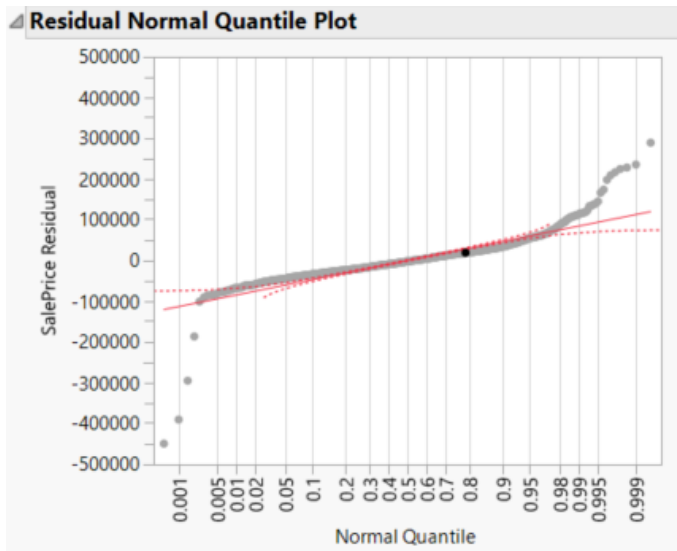


Image A: House at 3322 Harrison Rd



Image B: 95% Confidence and Prediction Intervals for Sale Price of 3322 Harrison Rd.

13/0 Cells											
Σ	Overall_Qual	Year_Built	First_FlrSF	GRLivArea	Fireplaces	GarageArea	SalePrice	Lower 95% Mean SalePrice 3	Upper 95% Mean SalePrice 3	Lower 95% Indiv SalePrice 3	Upper 95% Indiv SalePrice 3
2051	7	2004	2027	3379	1	848	•	346232.25657	360692.78681	282628.54465	424296.49873